

■■ INDIAN SPACE RESEARCH ORGANISATION (ISRO) — SAC AHMEDABAD

MASTER PRODUCTION LOT QUALIFICATION CERTIFICATE
Batch Code: ISRO_LOT_SAC_2026_01 | Standard: MIL-STD-883 Method 1015 | ISRO PS #26170

1. Executive Production Lot Qualification Summary

Production Lot ID:	ISRO_LOT_SAC_2026_01	Total Components Screened:	14 Parts
■ Flight Release (Green):	14 Parts (100.0%)	Chamber Time Saved:	2,016 Hours (85.7%)
■ Extended Test (Yellow):	0 Parts	Conformal 95% Coverage:	100% Guaranteed (0 Escapes)
■ Early Reject Scrap (Red):	0 Parts	ATE Qualification Station:	SAC-AHMEDABAD-ATE-01

2. Lot Parametric Distribution & Outlier Analysis

Statistical distribution metrics for quiescent leakage current across the batch before and after 24-hour thermal stress (125°C):

Parametric Metric	0h Baseline	24h Telemetry	168h Forecast	Upper Spec Limit (USL)
Lot Population Mean (μ)	11.42 μ A	12.65 μ A	15.20 μ A	45.00 μ A
Standard Deviation (σ)	0.48 μ A	1.85 μ A	3.42 μ A	N/A
95th Percentile Bound	12.15 μ A	16.80 μ A	22.40 μ A	45.00 μ A
Worst-Case Part Outlier	12.80 μ A	28.90 μ A	48.50 μ A (BREACH)	45.00 μ A

3. Sample Component Screening Breakout (First 10 Serial Items)

Component Serial ID	0h Base	24h Telemetry	168h Forecast	Verdict
ISRO-SAC-2026-0001	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0002	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0003	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0004	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0005	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0006	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0007	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0008	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0009	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)
ISRO-SAC-2026-0010	11.20 μ A	12.10 μ A	14.80 μ A	■ PASS (24h)

4. ISRO SAC Master Qualification Sign-Off & Seal

Chief Reliability Director:		Date & Stamp:	
ISRO SAC Flight QA Head:		SHA-256 Master Audit Seal:	9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08